

Advanced Python and Machine Learning

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5th / 6th / 12th / 13th May 2024

Please:



+



for class;

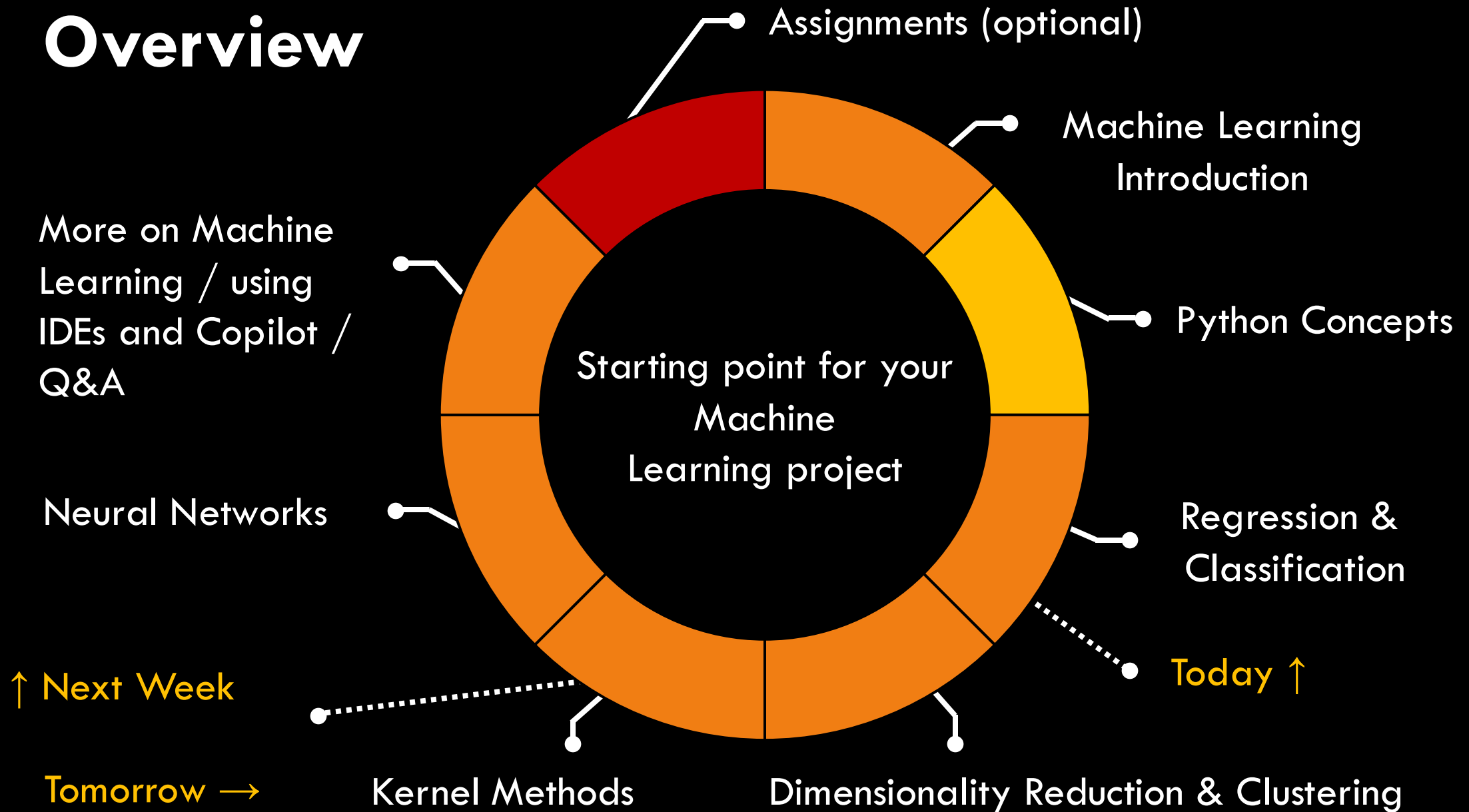


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for questions

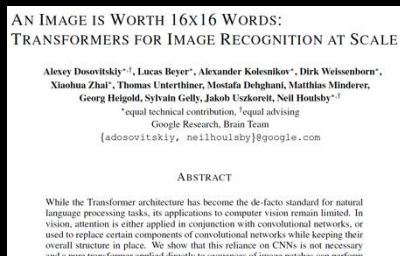
Overview



Practical Exercises

- End-of-notebook exercises → **breakout rooms of 2-3 people**
 - Goal: Understanding essentials to use different tools
 - Comparison of different methods for the same regression problem
- **Optional** assignments for more practice
 - 1 ECTS for completing **3 of 4** (your choice) **fully and reasonably**
 - **Deadline: 15th of June**

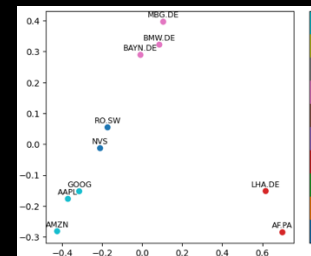
1. Parsing a PDF in Python +
generators + decorators



2. Ear-shells data + ridge regression +
decision trees + random forests



3. Finance data + PCA +
k-means

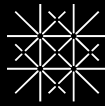


4. Fashion images +
neural networks



ETH zürich

ENE
ENS DE LYON

 University
of Basel

 **SDSC**

B.Sc. and M.Sc. in Physics

Ph.D. in Machine Learning

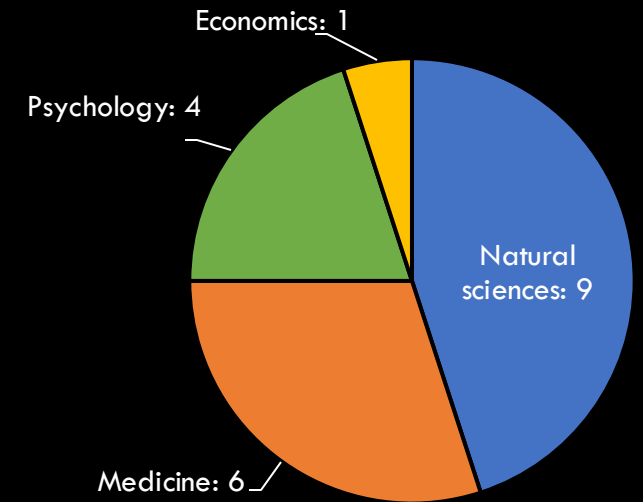
Research in Generative AI, Computer Vision,
Deep Learning theory and various applications

Senior Data Scientist and ML / AI researcher
Projects on climate modelling and drug safety



Your Experience and Goal

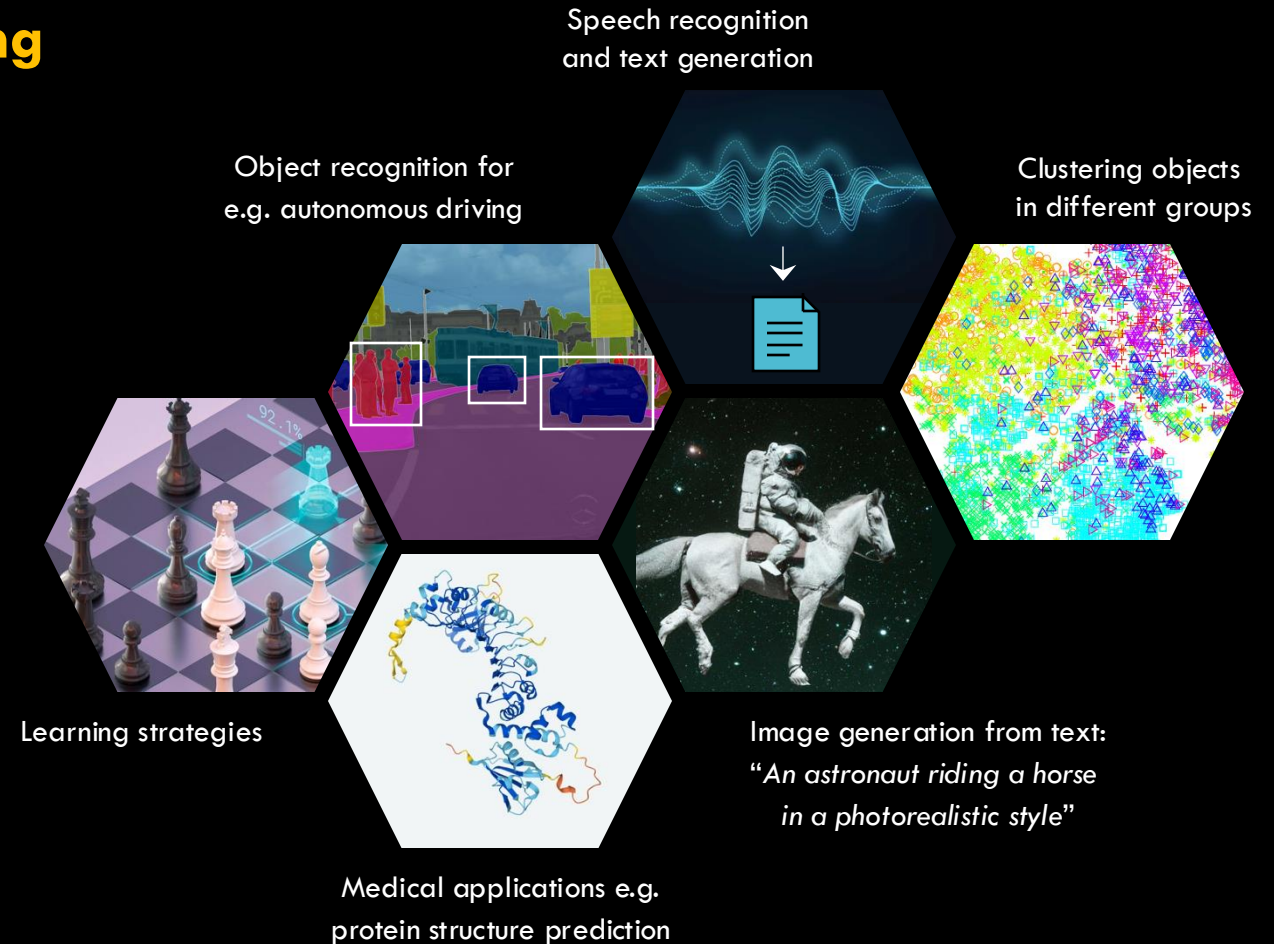
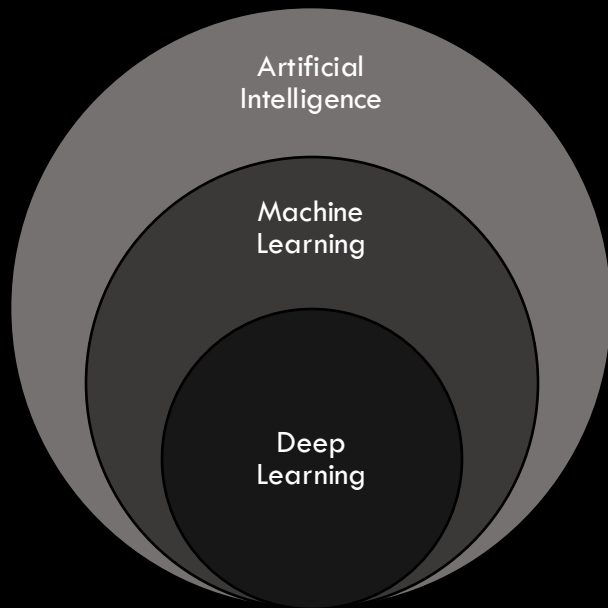
- What is your **ML / programming experience** so far and an **intermediate goal** for you?



Sena → Timon → Volodimir → Hélène → Ghéreint →
→ Mano → Linus → Fabian J. → Malwina → Cornelius →
→ Jan → Susanne → Fabian S. → Roman → Jose →
→ Thorald → Alessandro → Marten → Heiner → Sera →
→ Marietta → Eric

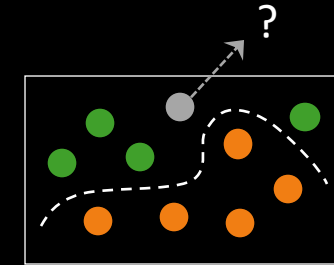
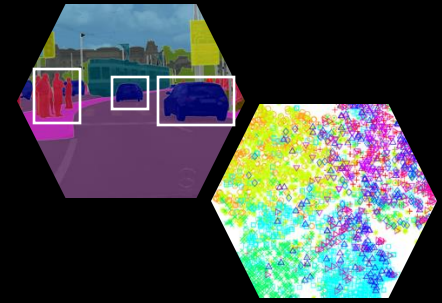
Machine Learning Landscape

- **Supervised** and **unsupervised learning**
- Discriminative and generative models
- **Reinforcement learning**

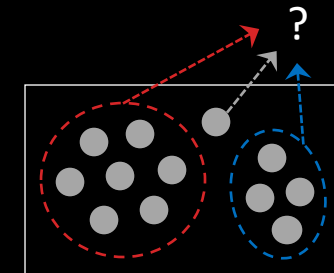


Supervised vs. Unsupervised

- **Supervised learning:** Labeled data guides learning
 - **Classification:** Class labels separate groups of data
 - **Regression:** Functional relationship between inputs (predictors) x and responses $y \rightarrow f(x) = y$
- **Unsupervised learning:** No labels available
 - **Clustering:** Identifying clusters inherent to the data
 - **Dimensionality reduction:** Identifying relevant predictors / dimensions or factors of variation in data



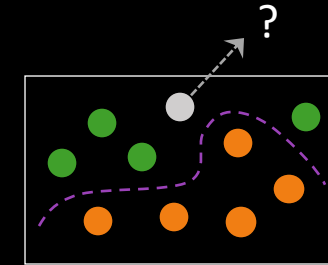
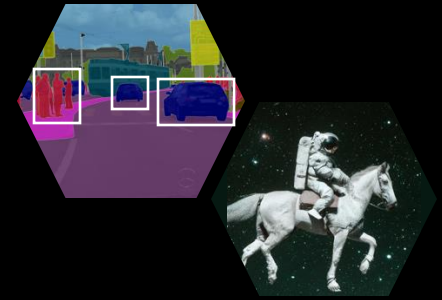
Supervised



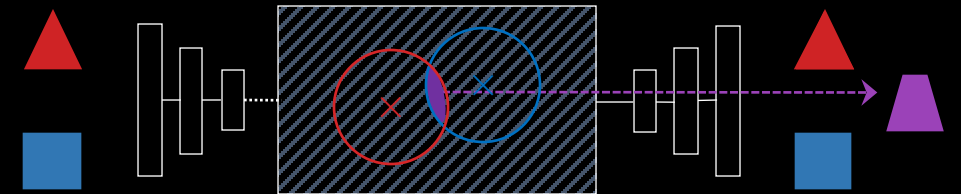
Unsupervised

Discriminative vs. Generative

- **Discriminative models:** Learn **decision boundary** to differentiate data and provide accurate predictions to new, unseen data
- **Generative models:** Approximate data distribution to provide accurate predictions but also allow **generating artificial data** similar to training data



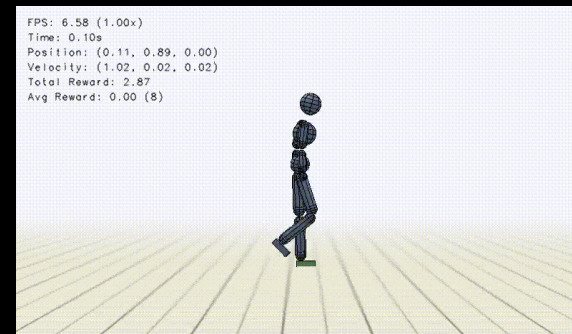
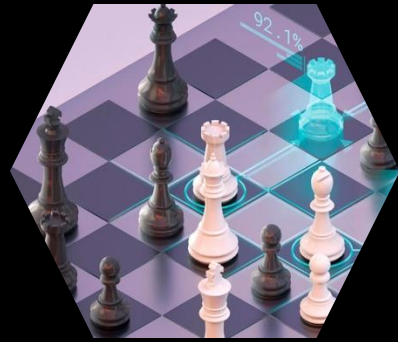
Discriminative



Generative

Reinforcement Learning

- Agents explore environment, select actions, receive rewards, and form strategies / policies → **exploration** and **exploitation**
- Modelling (Markov) decision processes
- Challenging in (large) uncertain environments
- Application in games, robotic, and finance

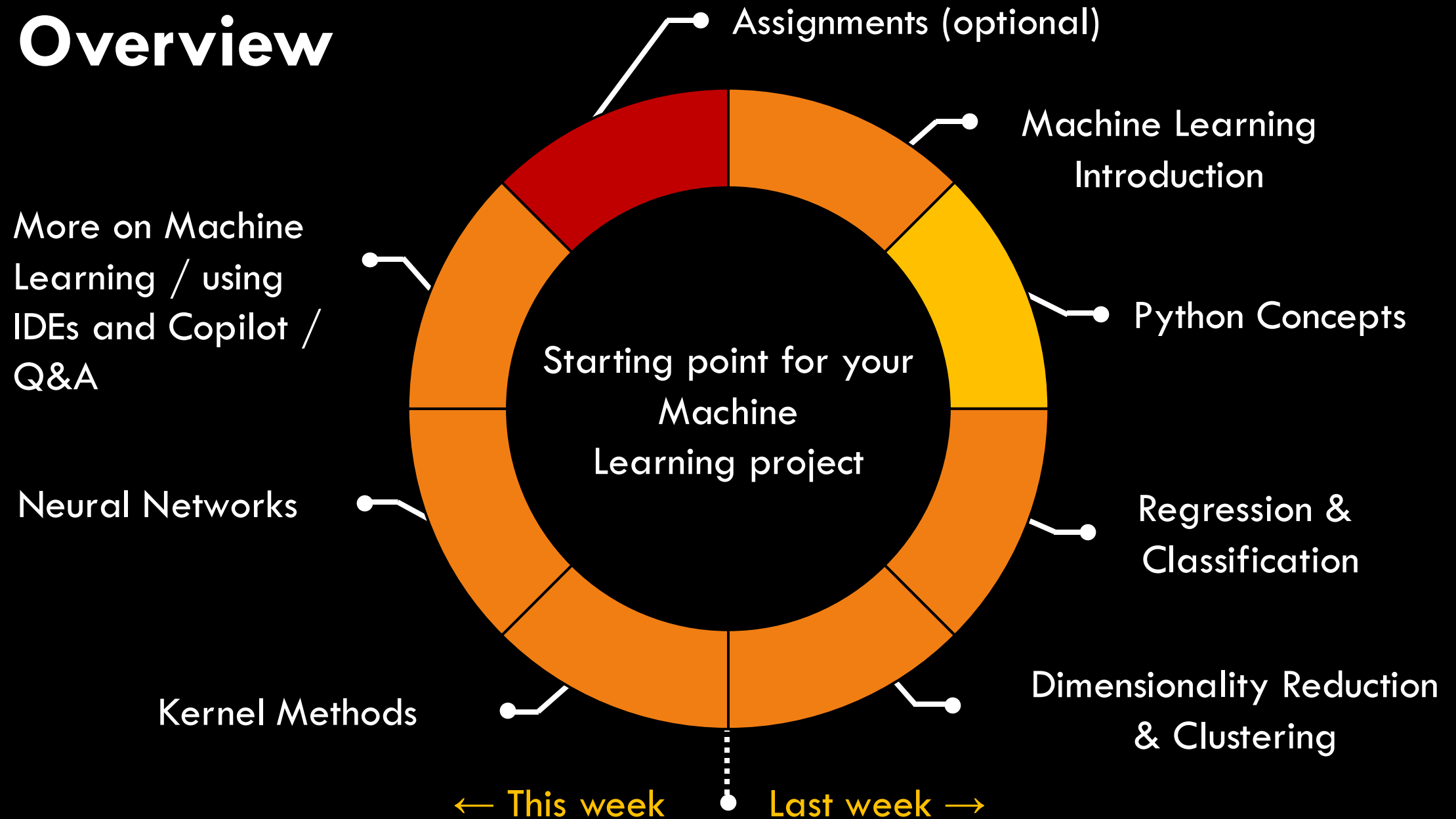


Source: [G. Berseth](#) and [C. Pal](#)

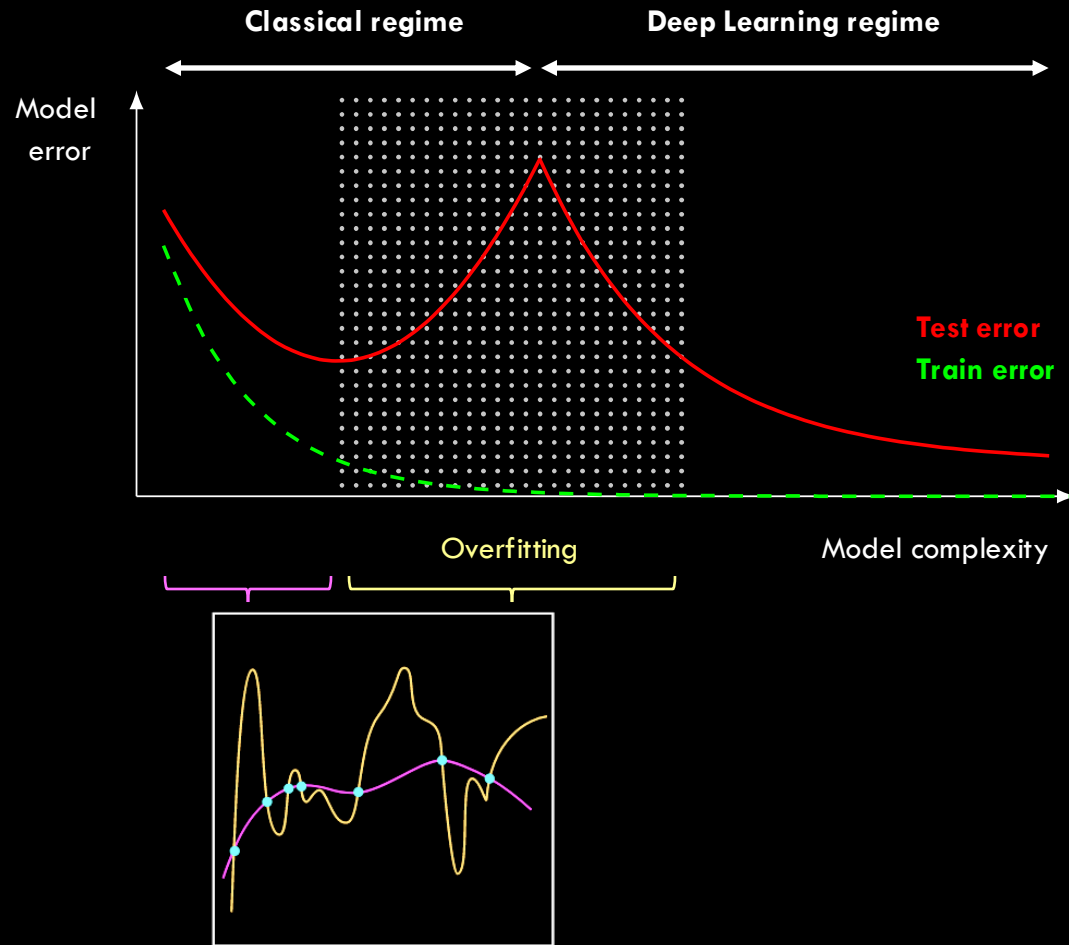
Start your Working Environment

1. Access the course environment via **Renku link**, sign in with your SWITCH edu-ID and launch the session
 2. **Or:** Download new material, start Jupyter Lab and open notebooks
- Suggestions on how to work in this course:
 -  +  : Follow presentation, while executing scripts yourself, making adjustments and notes in your own notebook
 -  : Follow presentation, switch to programming environment for exercises

Overview



Bias-Variance Trade-off



- Flexible models are prone to overfitting
- **Training / validation / test** splits to ensure generalisation to new data
- Watch out for imbalance in training data

Machine Learning Topics

Regression & Classification

Dim. Reduction & Clustering

Kernel Methods

Neural Networks

Linear / Ridge / Logistic Regression

Decision Trees & Random Forests

k-Nearest Neighbors

Support Vector Machines

Naïve Bayes

...

Principal Component Analysis

k-Means

Gaussian Mixture Models

Factor Analysis

Independent Component Analysis

...

Kernel Ridge Regression

Gaussian Process

Kernel Density Estimation

Kernel PCA

Kernel SVM

Kernel ...

Feed-Forward / Convolutional Neural Networks

Variational Autoencoder

Generative Adversarial Networks

Transformer

Diffusion Probabilistic Models

...

How to Continue with ML & Python

- Optional assignments: **Deadline 15th June**, 1 ECTS for **3 of 4 completed**
- Machine Learning **scikit-learn** user guide
https://scikit-learn.org/stable/user_guide.html
- **TensorFlow** tutorials
<https://www.tensorflow.org/resources/learn-ml>
- **Quick overview** with interactive tutorials on some basic topics and more advanced concepts:
<https://www.learnpython.org/>
- **Great tutorials** on specific (advanced) topics, easy-to-read books:
<https://www.realpython.com/>

How to Continue Programming

- **Visual Studio Code** (IDE): Free for all
 - Installation: <https://code.visualstudio.com/>
- **PyCharm Professional** (IDE): Free for students (Ph.D.) & teachers (PostDoc)
 - Installation: <https://www.jetbrains.com/pycharm/>
 - License: <https://www.jetbrains.com/community/education/#students>
- **GitHub Copilot** Pro: Free for students (Ph.D.) & teachers (PostDoc)
 - Get a certificate of employment ("Arbeitsbestätigung") from HR (in EN)
 - Documentation for license: <https://docs.github.com/en/copilot/managing-copilot/managing-copilot-as-an-individual-subscriber/managing-your-github-copilot-pro-subscription/getting-free-access-to-copilot-pro-as-a-student-teacher-or-maintainer>
- Use high-performance computing (GPUs, TBs of RAM) cluster **sciCORE**
 - **Getting started:**
<https://scicore.unibas.ch/using-scicore/getting-started/>
 - Courses:
<https://scicore.unibas.ch/events/teaching-training/>

Suggestions for the Feedback



- **Preparation information** / YouTube video adequate?
- What did / didn't you like about **Renku**?
- Content appropriate: **anything missing** for you (Python or Machine Learning)?
- Too fast or slow, shallow or deep?
- **Insightful exercises** and enough time?

Thank you and good luck on your Machine Learning journey! 😊